

Conte Operacon QR Givens

Aplacar rotación a A

$$\begin{bmatrix} c & s \\ -s & c \end{bmatrix} A_{j:j} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \begin{bmatrix} a_{jj} & a_{jj+1} & \dots & a_{jn} \\ a_{ij} & a_{ij+1} & \dots & a_{in} \end{bmatrix}$$

$$\begin{bmatrix} c \cdot [a_{jj} \dots a_{jn}] + s[a_{ij} \dots a_{in}] \\ -s[a_{jj} \dots a_{jn}] + c[a_{ij} \dots a_{in}] \end{bmatrix} \quad \begin{matrix} 6(n-j+1) \\ \text{flops} \end{matrix}$$

Aplicar rotación traspuesta a Q

$6m$ flops

$$\begin{bmatrix} | & | \\ \textcircled{Q} & \\ | & | \end{bmatrix} \begin{bmatrix} 6^{\uparrow} \end{bmatrix}$$

fila j fila i generas C y S

Generación de R :

$$\sum_{j=1}^p \sum_{i=j+1}^m (\textcircled{Q} + 6(n-j+1))$$

Generación de Q :

$$\sum_{j=1}^p \sum_{i=j+1}^m 6m$$

$$\sum_{j=1}^p \sum_{i=j+1}^m (6 + 6(n-j+1)) = \sum_{j=1}^p \sum_{i=j+1}^m 6(n-j+2) =$$

$$= \sum_{j=1}^p 6(n-j+2)(m-j) =$$

$$= 3p^2(m+n - \frac{4}{3}p) + 6p(m-p-1)(n-p) +$$

$$+ 9p(m-p) + 3pn - 5p$$

$$= O\left(3p^2(m+n - \frac{4}{3}p)\right)$$

$$\boxed{O(n^3)}$$

$$\text{So } m=n \Rightarrow p \sim n, \quad O\left(3n^2(n+n - \frac{4}{3}n)\right)$$

$$Q: \sum_{j=1}^p \sum_{i=j+1}^m 6m = \sum_{j=1}^p 6m(m-j) =$$

$$= 6mp\left(m - \frac{1}{2}p\right) - 3mp$$

$$\bigcirc \left(6mp\left(m - \frac{1}{2}p\right)\right)$$

$$Q_u = I - \frac{2}{\|u\|_2^2} u u^T = "Q"$$

$$i) Q u = \left(I - \frac{2}{\|u\|_2^2} u u^T \right) u = u - \frac{2}{\|u\|_2^2} u (u^T u)$$

$$= u - 2u = -u$$

$$ii) v \perp u \Rightarrow \left(I - \frac{2}{\|u\|_2^2} u u^T \right) v =$$

$$= v - \frac{2}{\|u\|_2^2} u (u^T v) = v$$

$$\text{iii) } Q^T = \left(I - \frac{2}{\|u\|_2^2} u u^T \right)^T = I^T - \frac{2}{\|u\|_2^2} (u u^T)^T$$

$$= I - \frac{2}{\|u\|_2^2} u u^T = Q$$

$$\text{iv) } Q Q = \left(I - \frac{2}{\|u\|_2^2} u u^T \right) \left(I - \frac{2}{\|u\|_2^2} u u^T \right) =$$

$$= I - \frac{4}{\|u\|_2^2} u u^T + \frac{4}{\|u\|_2^2} u (u^T u) u^T = I$$